

# NanoGT Match Report: Mucopolysaccharidosis type II

**Disease:** Mucopolysaccharidosis type II (ORPHA:580)

**Primary gene:** IDS

**Gene CDS:** 1650 bp

**Inheritance:** X-linked recessive

**Target tissues:** CNS, liver

## Top 5 GT Precedent Matches

| Rank | Program   | Vector | Score  | Confidence   | Approval |
|------|-----------|--------|--------|--------------|----------|
| 1    | Libmeldy  | LV     | 9.2/10 | [GREEN] High | approved |
| 2    | ST-920    | AAV2/6 | 8.8/10 | [GREEN] High | phase1/2 |
| 3    | RGX-121   | AAV9   | 8.7/10 | [GREEN] High | phase3   |
| 4    | Hemgenix  | AAV5   | 8.6/10 | [GREEN] High | approved |
| 5    | Roctavian | AAV5   | 8.6/10 | [GREEN] High | approved |

## Match #1: Libmeldy

**Precedent disease:** Metachromatic leukodystrophy

**Vector:** LV

**Tissue target:** hematopoietic/CNS

**Composite score:** 9.2 / 10

## Score Breakdown

| Dimension                 | Score | Max | What it measures                                     |
|---------------------------|-------|-----|------------------------------------------------------|
| Packaging fit             | 2.00  | 2.0 | Gene CDS size vs vector cargo capacity               |
| Tissue tropism            | 1.50  | 2.0 | Vector naturally reaches disease target tissue       |
| Protein class             | 2.00  | 2.0 | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 2.00  | 2.0 | Same or related biological pathway                   |
| Inheritance compatibility | 0.70  | 1.0 | AR/XL loss-of-function pattern match                 |
| Approval precedent        | 1.00  | 1.0 | Regulatory approval / trial stage                    |
| Immunogenicity            | 2.00  | 2.0 | Pre-existing NAb seroprevalence for this vector      |
| Therapeutic window        | 1.50  | 2.0 | Can GT be given before irreversible damage?          |

| Dimension                 | Score       | Max         | What it measures                                     |
|---------------------------|-------------|-------------|------------------------------------------------------|
| Cross-correction          | 1.00        | 1.0         | Can transduced cells rescue untransduced neighbours? |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue            |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist            |
| Route of administration   | 1.00        | 1.0         | Established delivery route to target tissue          |
| <b>TOTAL (normalised)</b> | <b>9.22</b> | <b>10.0</b> | Raw sum / $18 \times 10$                             |

## Rationale

- Gene CDS 1650bp / cargo 8000bp (21% utilized)
- Precedent target match: cns
- Both lysosomal proteins — cross-correction likely
- LOF inheritance — compatible for gene replacement
- Pathway match: leukodystrophy
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/haAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

## Match #2: ST-920

**Precedent disease:** Fabry disease

**Vector:** AAV2/6

**Tissue target:** liver

**Composite score:** 8.8 / 10

## Score Breakdown

| Dimension      | Score | Max | What it measures                               |
|----------------|-------|-----|------------------------------------------------|
| Packaging fit  | 1.50  | 2.0 | Gene CDS size vs vector cargo capacity         |
| Tissue tropism | 2.00  | 2.0 | Vector naturally reaches disease target tissue |

| Dimension                 | Score       | Max         | What it measures                                      |
|---------------------------|-------------|-------------|-------------------------------------------------------|
| Protein class             | 2.00        | 2.0         | Same se-creted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 2.00        | 2.0         | Same or related biological pathway                    |
| Inheritance compatibility | 1.00        | 1.0         | AR/XL loss-of-function pattern match                  |
| Approval precedent        | 0.50        | 1.0         | Regulatory approval / trial stage                     |
| Immunogenicity            | 1.50        | 2.0         | Pre-existing NAb seroprevalence for this vector       |
| Therapeutic window        | 1.50        | 2.0         | Can GT be given before irreversible damage?           |
| Cross-correction          | 1.00        | 1.0         | Can transduced cells rescue untransduced neighbours?  |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue             |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist             |
| Route of administration   | 1.00        | 1.0         | Established delivery route to target tissue           |
| <b>TOTAL (normalised)</b> | <b>8.83</b> | <b>10.0</b> | Raw sum / 18 × 10                                     |

## Rationale

- Gene CDS 1650bp / cargo 4700bp (35% utilized)
- Vector tropism plus precedent target match: liver
- Both lysosomal proteins — cross-correction likely
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: lysosomal\_storage
- Approval status: phase1/2
- Vector immunogenicity (AAV2/6): moderate (~17%) — significant proportion may require pre-screening or exclusion
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

## Match #3: RGX-121

**Precedent disease:** Mucopolysaccharidosis type II

**Vector:** AAV9

**Tissue target:** CNS/liver  
**Composite score:** 8.7 / 10

### Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                      |
|---------------------------|-------------|-------------|-------------------------------------------------------|
| Packaging fit             | 1.50        | 2.0         | Gene CDS size vs vector cargo capacity                |
| Tissue tropism            | 2.00        | 2.0         | Vector naturally reaches disease target tissue        |
| Protein class             | 2.00        | 2.0         | Same se-creted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 2.00        | 2.0         | Same or related biological pathway                    |
| Inheritance compatibility | 1.00        | 1.0         | AR/XL loss-of-function pattern match                  |
| Approval precedent        | 0.80        | 1.0         | Regulatory approval / trial stage                     |
| Immunogenicity            | 1.00        | 2.0         | Pre-existing NAb seroprevalence for this vector       |
| Therapeutic window        | 1.50        | 2.0         | Can GT be given before irreversible damage?           |
| Cross-correction          | 1.00        | 1.0         | Can transduced cells rescue untransduced neighbours?  |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue             |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist             |
| Route of administration   | 1.00        | 1.0         | Established delivery route to target tissue           |
| <b>TOTAL (normalised)</b> | <b>8.72</b> | <b>10.0</b> | Raw sum / 18 × 10                                     |

### Rationale

- Gene CDS 1650bp / cargo 4700bp (35% utilized)
- Vector tropism plus precedent target match: cns, liver
- Both lysosomal proteins — cross-correction likely
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: lysosomal\_storage
- Approval status: phase3
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected

- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

#### Match #4: Hemgenix

**Precedent disease:** Hemophilia B

**Vector:** AAV5

**Tissue target:** liver

**Composite score:** 8.6 / 10

#### Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                     |
|---------------------------|-------------|-------------|------------------------------------------------------|
| Packaging fit             | 1.50        | 2.0         | Gene CDS size vs vector cargo capacity               |
| Tissue tropism            | 2.00        | 2.0         | Vector naturally reaches disease target tissue       |
| Protein class             | 2.00        | 2.0         | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 0.50        | 2.0         | Same or related biological pathway                   |
| Inheritance compatibility | 1.00        | 1.0         | AR/XL loss-of-function pattern match                 |
| Approval precedent        | 1.00        | 1.0         | Regulatory approval / trial stage                    |
| Immunogenicity            | 2.00        | 2.0         | Pre-existing NAb seroprevalence for this vector      |
| Therapeutic window        | 1.50        | 2.0         | Can GT be given before irreversible damage?          |
| Cross-correction          | 1.00        | 1.0         | Can transduced cells rescue untransduced neighbours? |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue            |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist            |
| Route of administration   | 1.00        | 1.0         | Established delivery route to target tissue          |
| <b>TOTAL (normalised)</b> | <b>8.56</b> | <b>10.0</b> | Raw sum / 18 × 10                                    |

#### Rationale

- Gene CDS 1650bp / cargo 4700bp (35% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Different pathway (lysosomal\_storage vs coagulation)

- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

## Match #5: Roctavian

**Precedent disease:** Hemophilia A

**Vector:** AAV5

**Tissue target:** liver

**Composite score:** 8.6 / 10

## Score Breakdown

| Dimension                 | Score | Max | What it measures                                     |
|---------------------------|-------|-----|------------------------------------------------------|
| Packaging fit             | 1.50  | 2.0 | Gene CDS size vs vector cargo capacity               |
| Tissue tropism            | 2.00  | 2.0 | Vector naturally reaches disease target tissue       |
| Protein class             | 2.00  | 2.0 | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 0.50  | 2.0 | Same or related biological pathway                   |
| Inheritance compatibility | 1.00  | 1.0 | AR/XL loss-of-function pattern match                 |
| Approval precedent        | 1.00  | 1.0 | Regulatory approval / trial stage                    |
| Immunogenicity            | 2.00  | 2.0 | Pre-existing NAb seroprevalence for this vector      |
| Therapeutic window        | 1.50  | 2.0 | Can GT be given before irreversible damage?          |
| Cross-correction          | 1.00  | 1.0 | Can transduced cells rescue untransduced neighbours? |
| Immune privilege          | 0.90  | 1.0 | Immunological protection of target tissue            |
| Promoter availability     | 1.00  | 1.0 | Validated tissue-specific promoters exist            |
| Route of administration   | 1.00  | 1.0 | Established delivery route to target tissue          |

| Dimension                     | Score       | Max         | What it measures  |
|-------------------------------|-------------|-------------|-------------------|
| <b>TOTAL<br/>(normalised)</b> | <b>8.56</b> | <b>10.0</b> | Raw sum / 18 × 10 |

## Rationale

- Gene CDS 1650bp / cargo 4700bp (35% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Different pathway (lysosomal\_storage vs coagulation)
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Moderate therapeutic window — progressive disease with childhood onset; early intervention strongly recommended; newborn screening integration beneficial
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs